

Appendix 14: Curve Plotting Functions

Instruction Manual

Instruction Manual Version: V2.04

Update Date: 2023.07.01

During CAN debugging, software can be used to parse bus data according to the protocol and plot corresponding physical quantity curves for analysis.

This function is described in detail below, using the CANPro software from Chou-Li-Gong.

1、Chou Li-Gong CANPro Software Use

When users use CANPro1.50, they only need to replace ControlCAN.dll and other library files (refer to: How to use Zhou Ligong CANpro1.50 software compatibly.pdf), and select the model: USBCAN-2E-U.

CANpro1.50 is more feature-rich, here to CANpro1.50 as an example.

Open CANpro1.50 software, select USBCAN-2E-U interface card, and selected bus baud rate (subject to the actual baud rate, automotive CAN bus is generally 500K), click OK and start, start the CAN interface card. As shown in Figure 1;

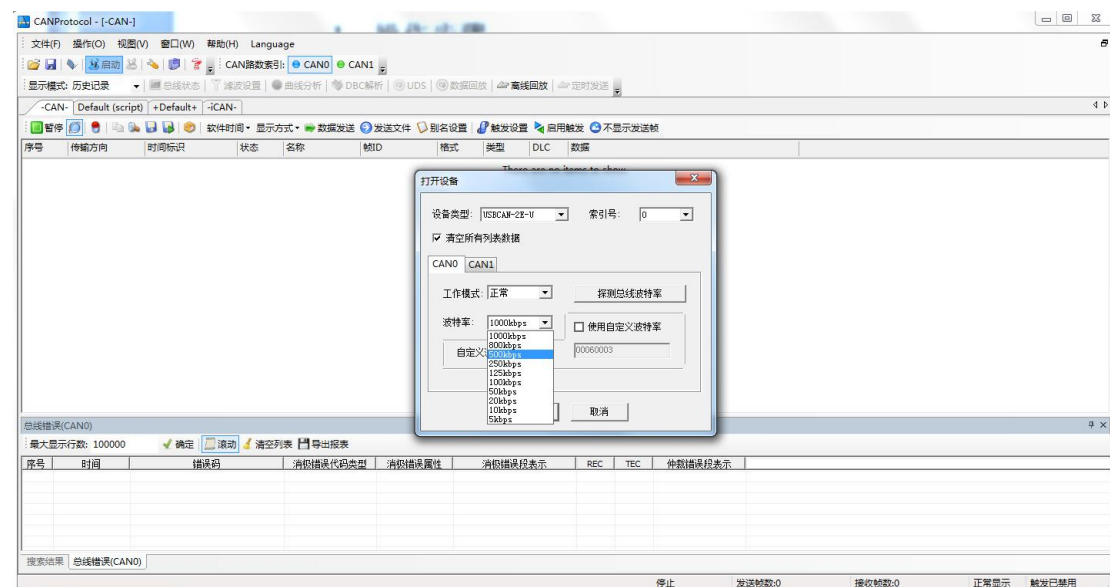
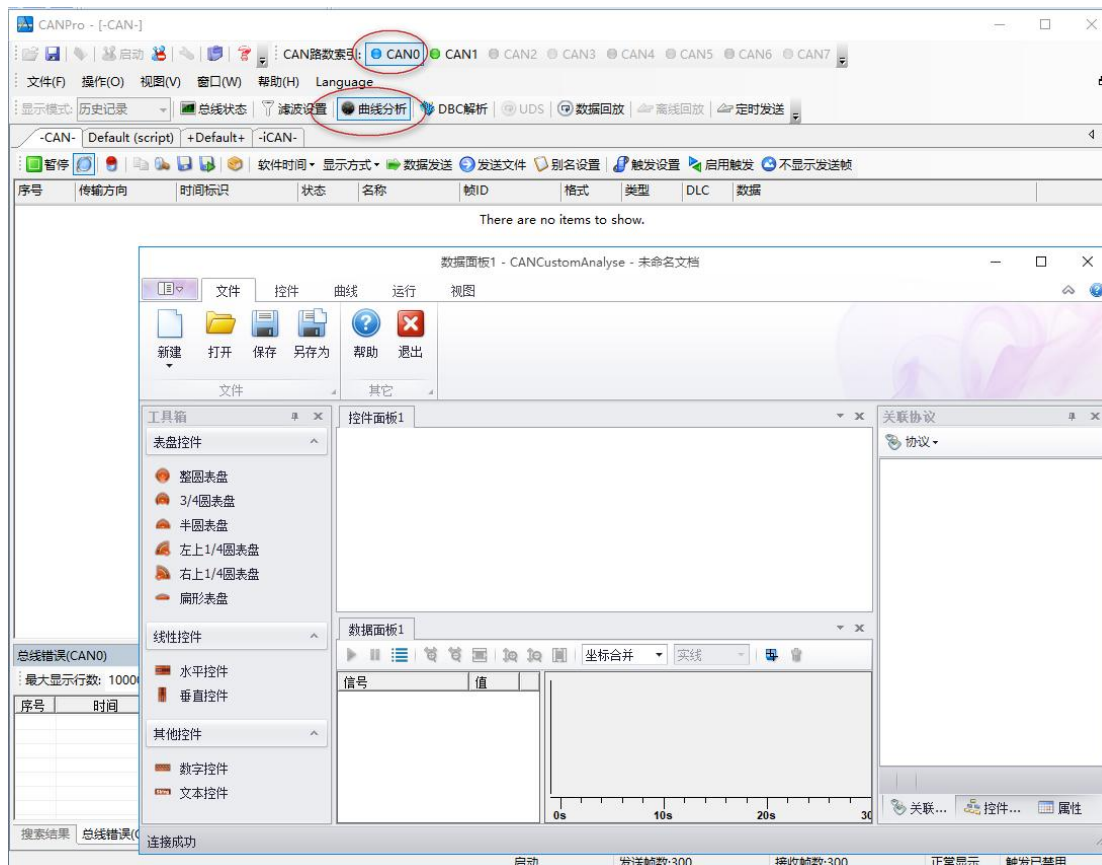


Fig. 1 Opening CANpro software initialisation

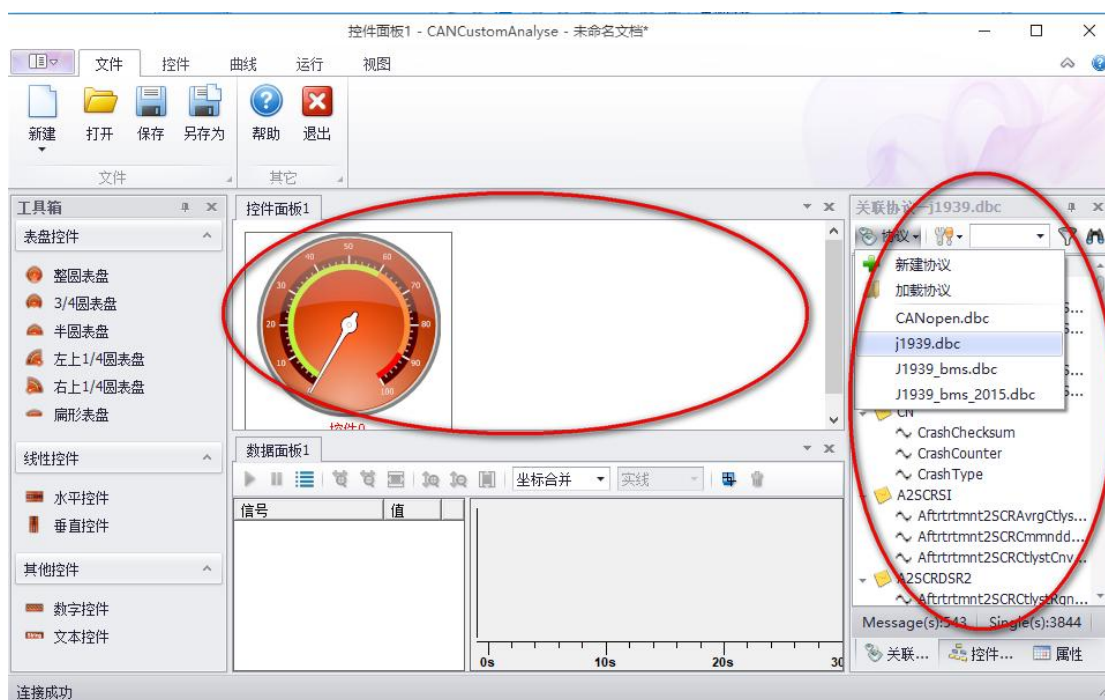
2. Load the Standard DBC Format Protocols

Click the "Curve Analysis" button in the shortcut menu to pop up the curve analysis interface. The parameters can be used without any modification.

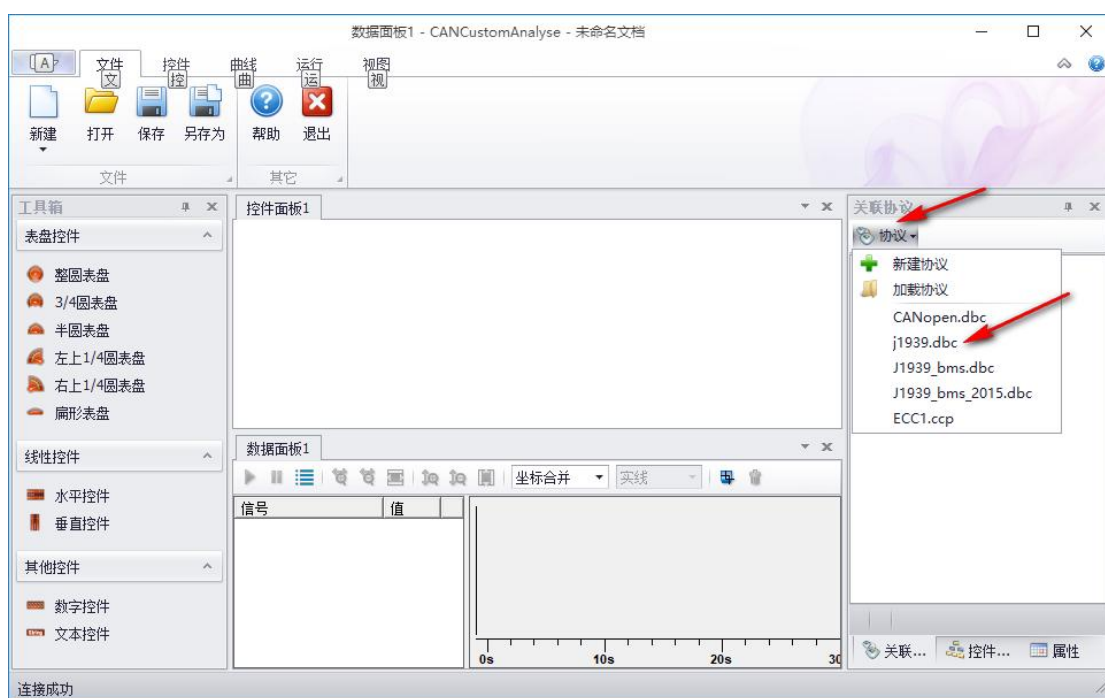


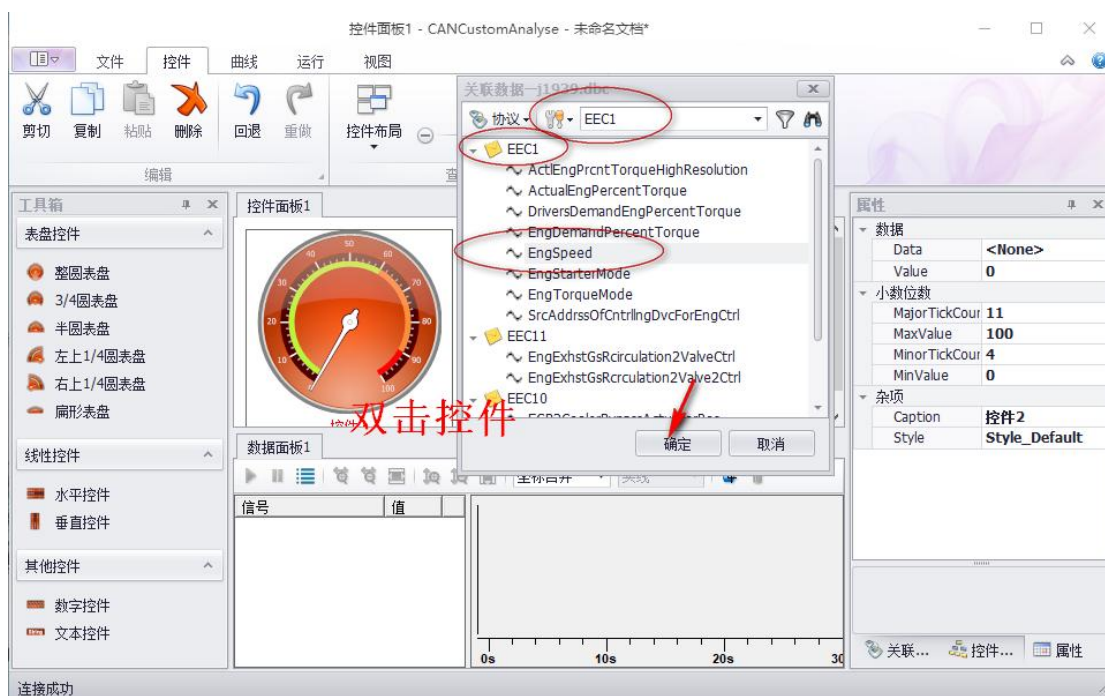
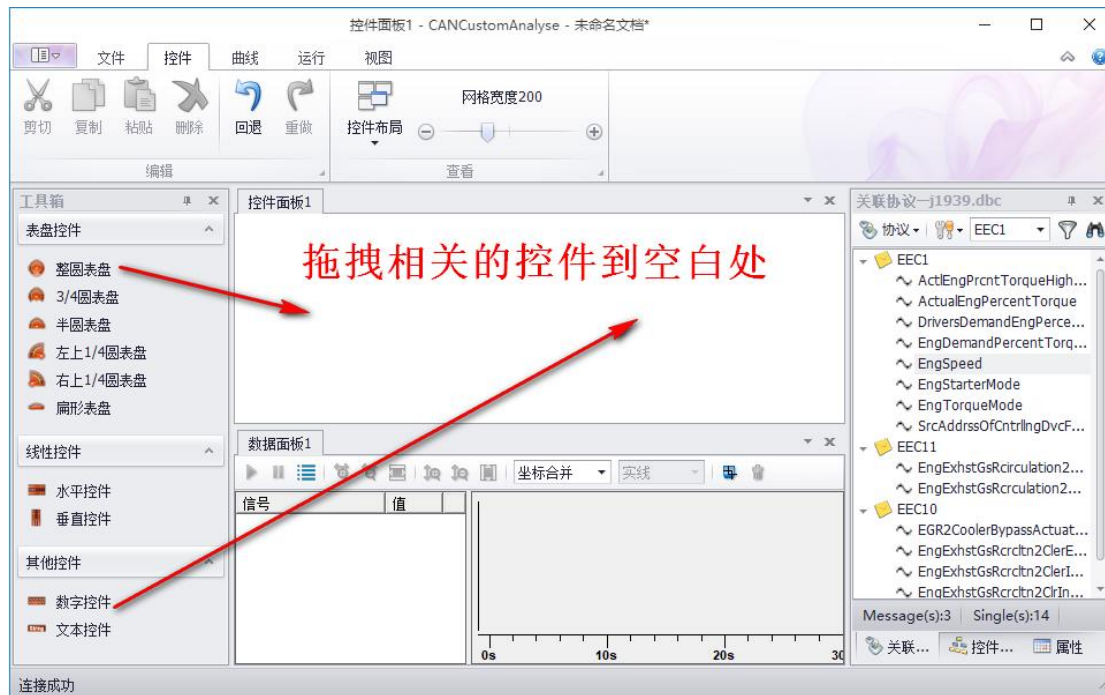
Note: CAN0 represents the first channel and CAN1 represents the second channel. Each channel has its own curve analysis interface. To analyse a specific channel, switch to the corresponding channel's data interface first.

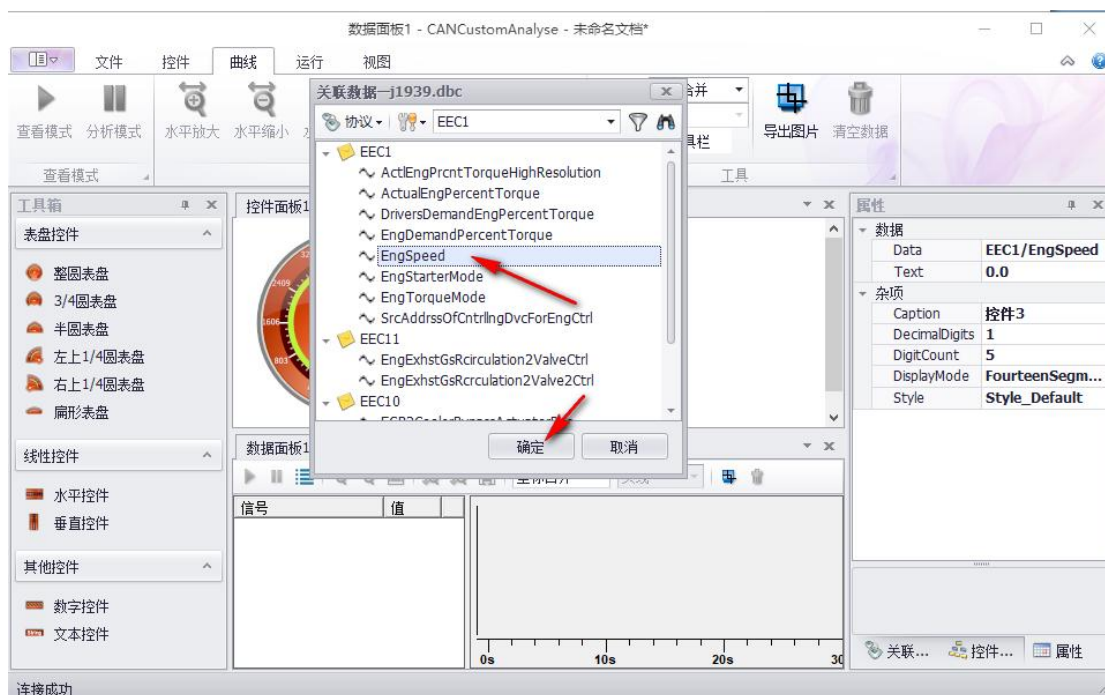
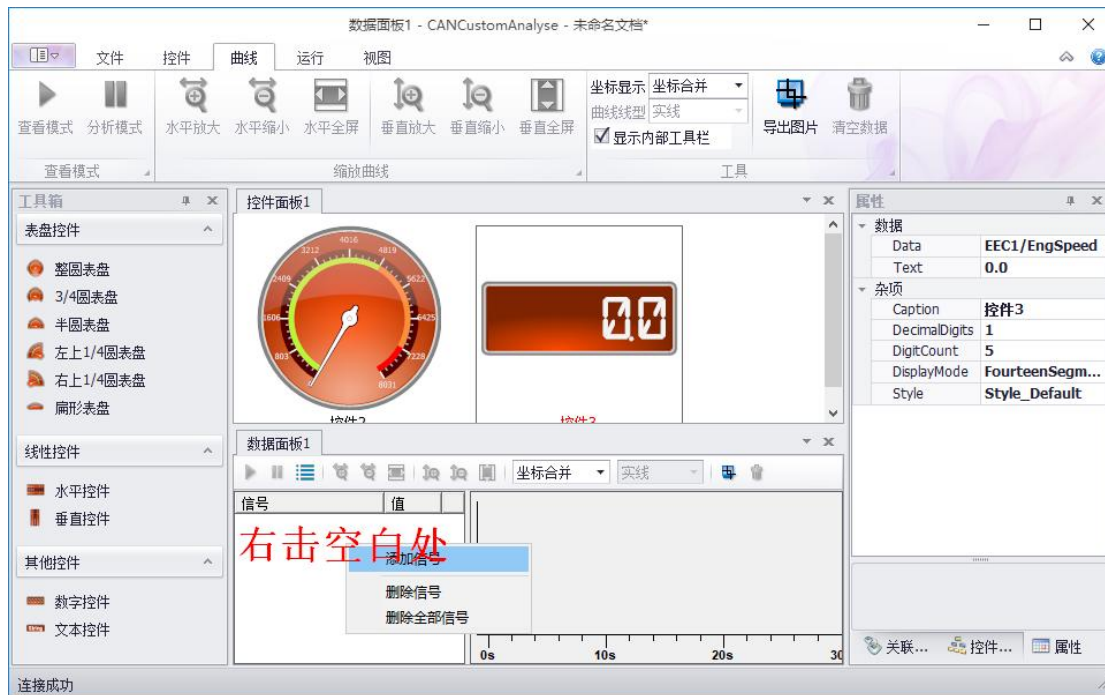
Curve analysis involves two main points: interface editing and protocol. The software offers standard protocols such as J1939, which can be directly loaded and parsed. However, for custom protocols, users must create their own 'new protocol'.

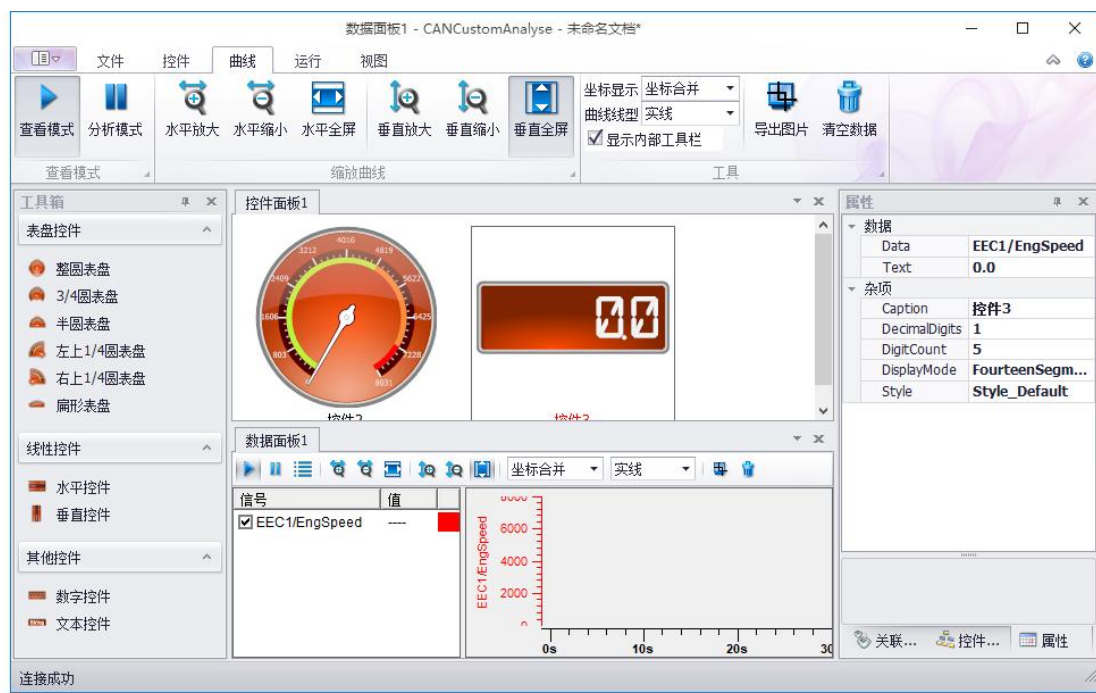


Load the standard J1939 protocol as instructed.

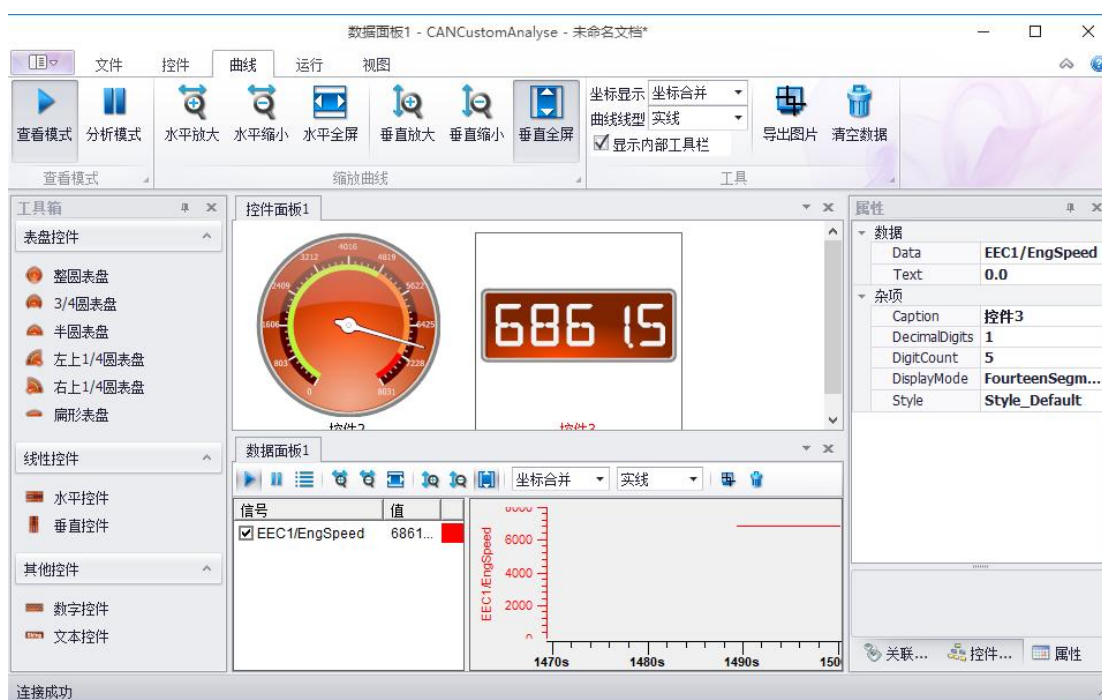








If the bus has received the data and the frame conforms to the protocol, it is then parsed according to the protocol.

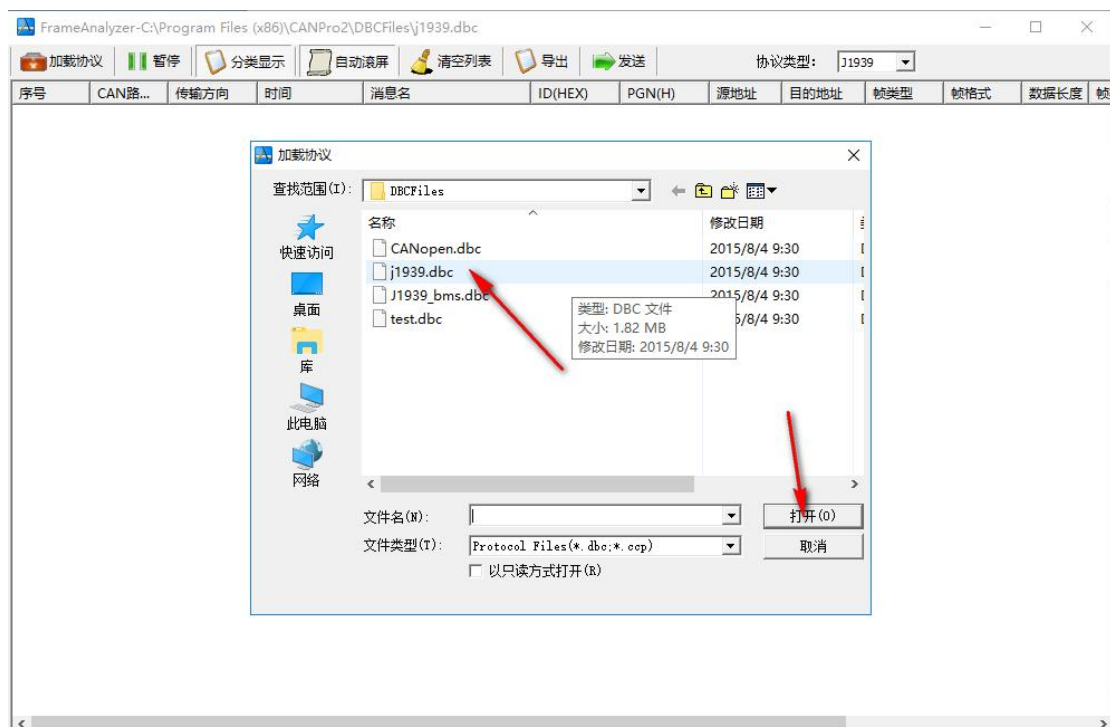
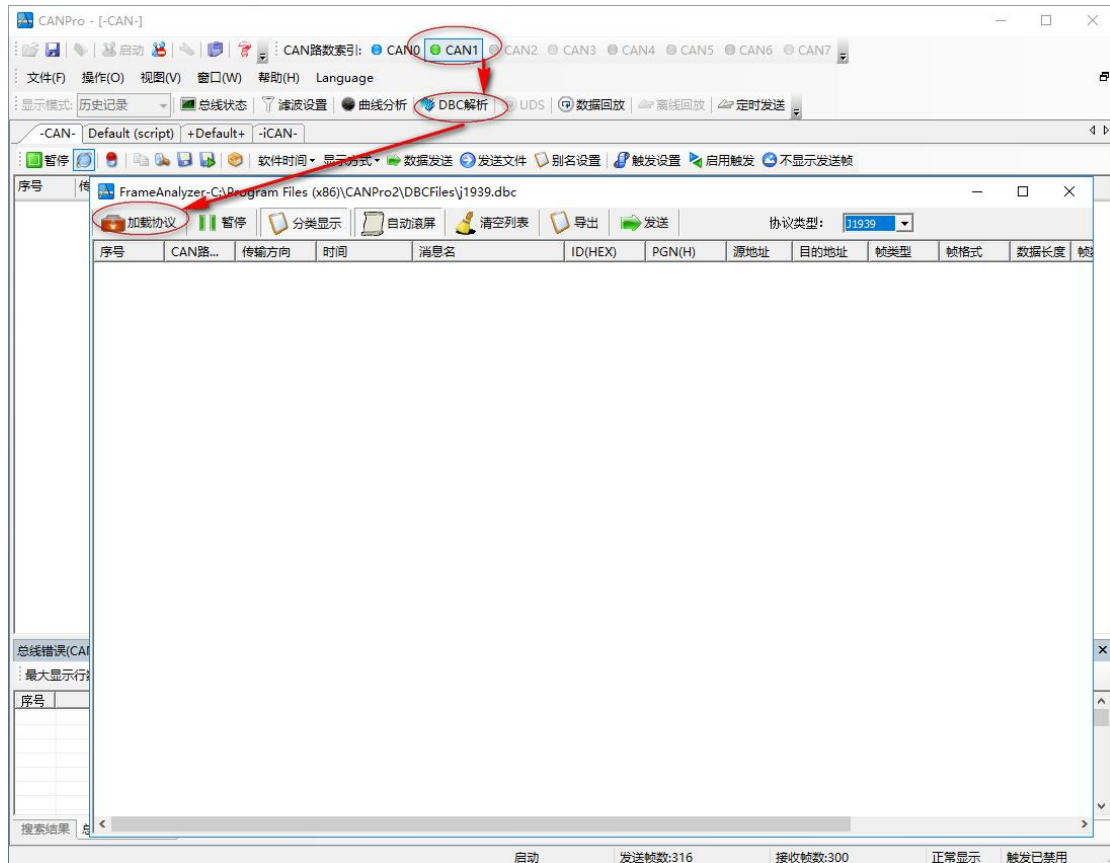


If there is no test equipment used to send the data source, CAN1, CAN2 can be shorted up to the pair of sending, one sends and one receives.

Wiring reference *"5. Plug-in 1: USB-CAN bus adapter test.pdf"*

The curve drawing above is opened in CAN0 interface, so it is used to parse the data

sent and received by CAN1. The following sending needs to be done under CAN1 interface.



2. Custom Protocol

For example, arbitrarily define the protocol for a frame of data:

ID=0x0CF0041A Extended frame Data frame Length of frame data 8 bytes

The 4th and 5th bytes represent EngSpeed, the other bytes are ignored.

Data length: 2 bytes

Resolution: 0.125 rpm/bit increment, counting from 0 rpm (high byte resolution = 32 rpm/bit)

Data range: 0 to 8031.875 rpm

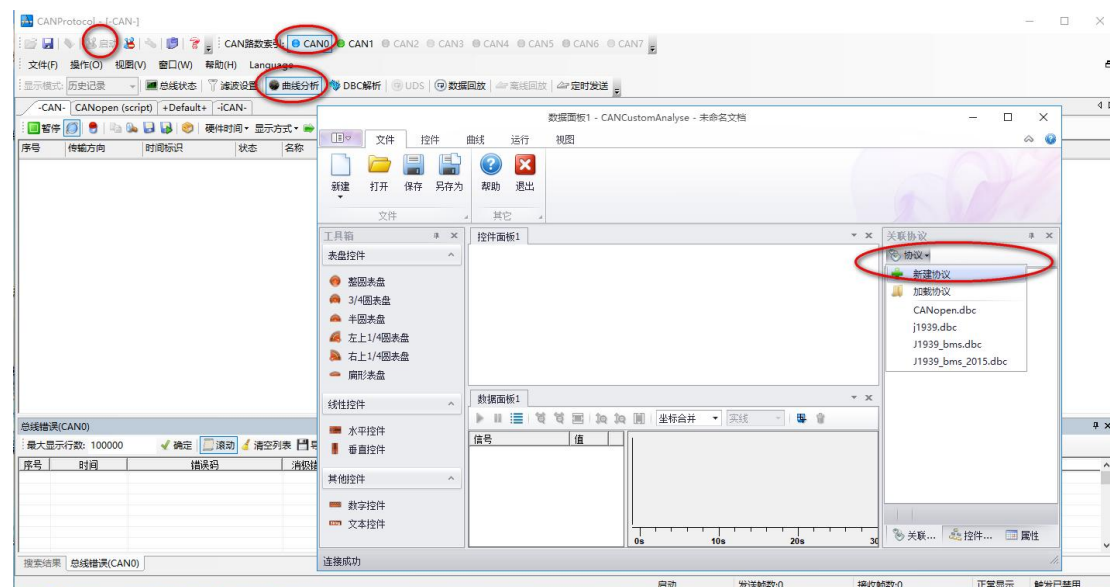
Assuming, frame received: ID = 0x0CF0041A, data 00 00 00 6C D6 00 00 00 00

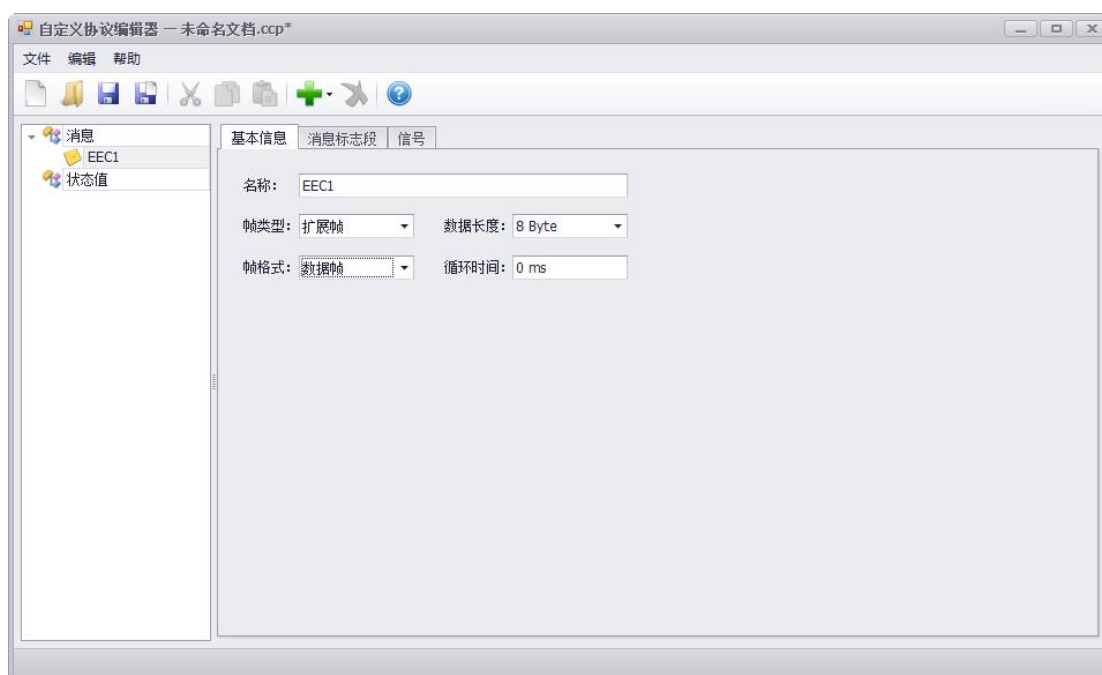
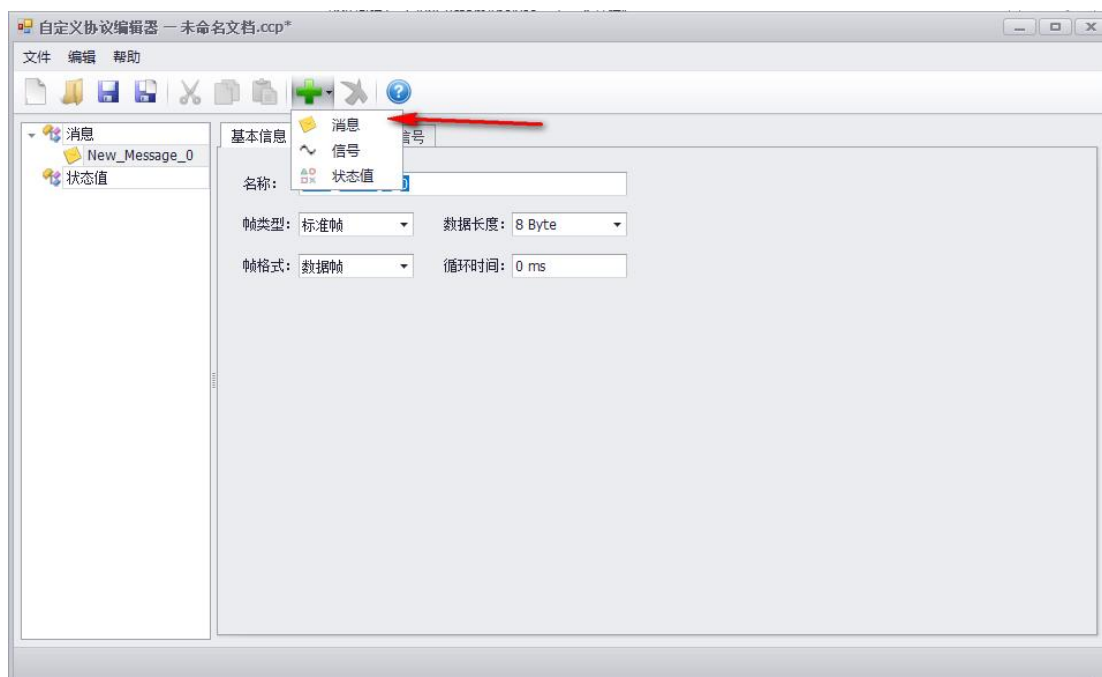
4th byte from left = 0x6C

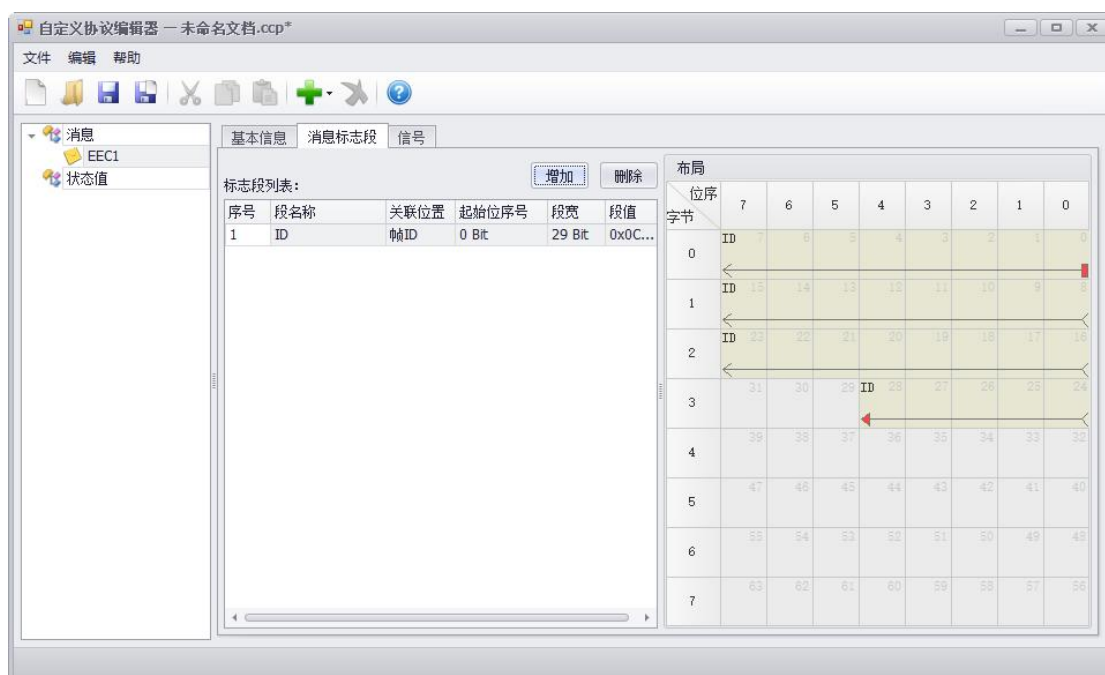
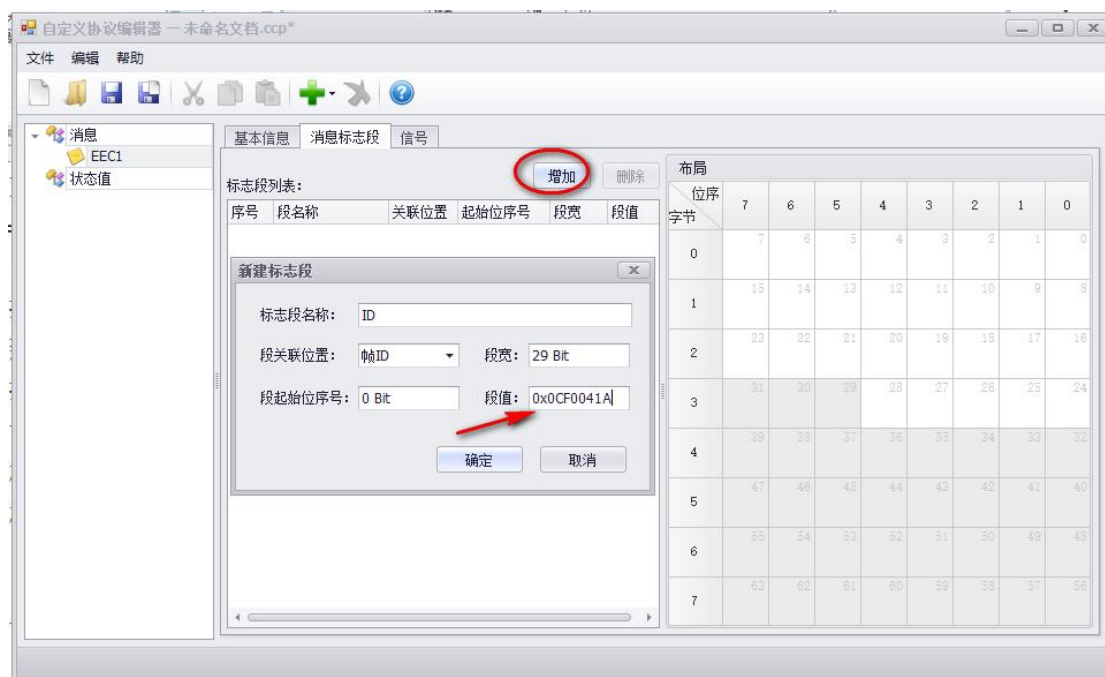
5th byte from left = 0xD6

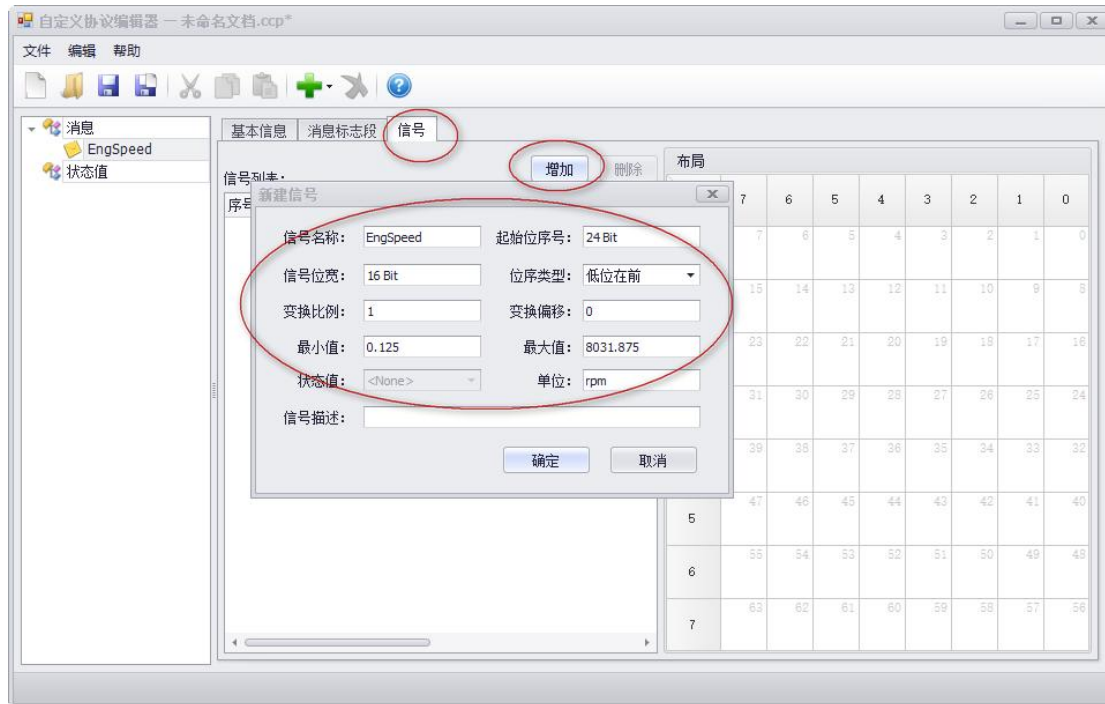
You can calculate the speed: $0xD66C \times 0.125$ is 6861.50rpm (revolutions per minute).

The following new protocol is created

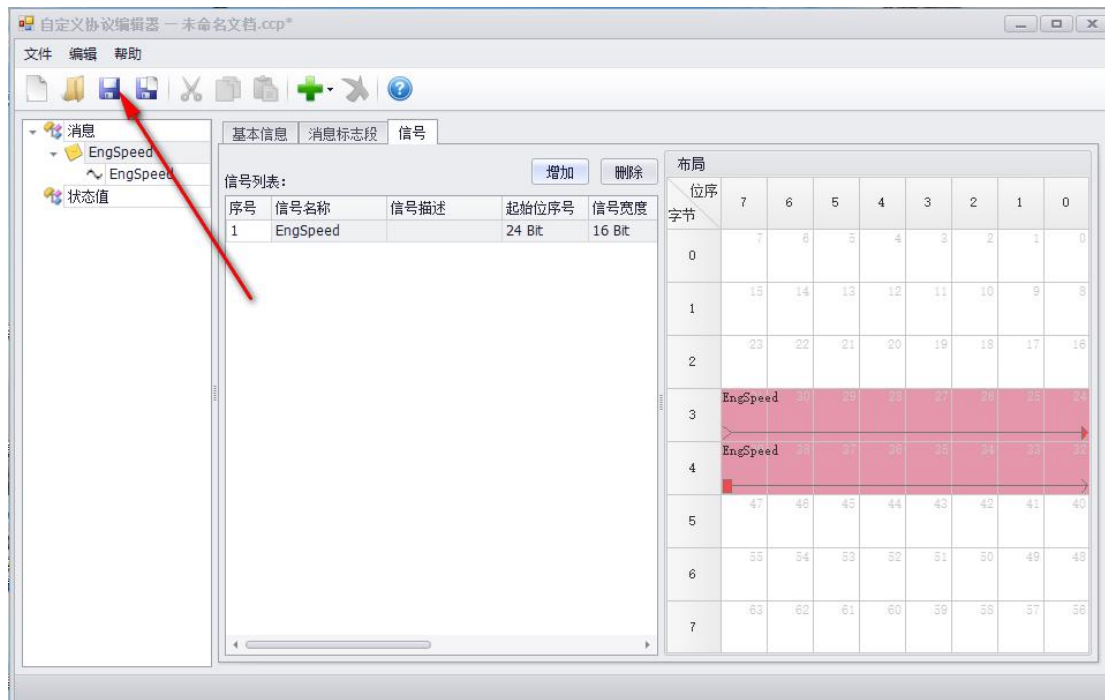


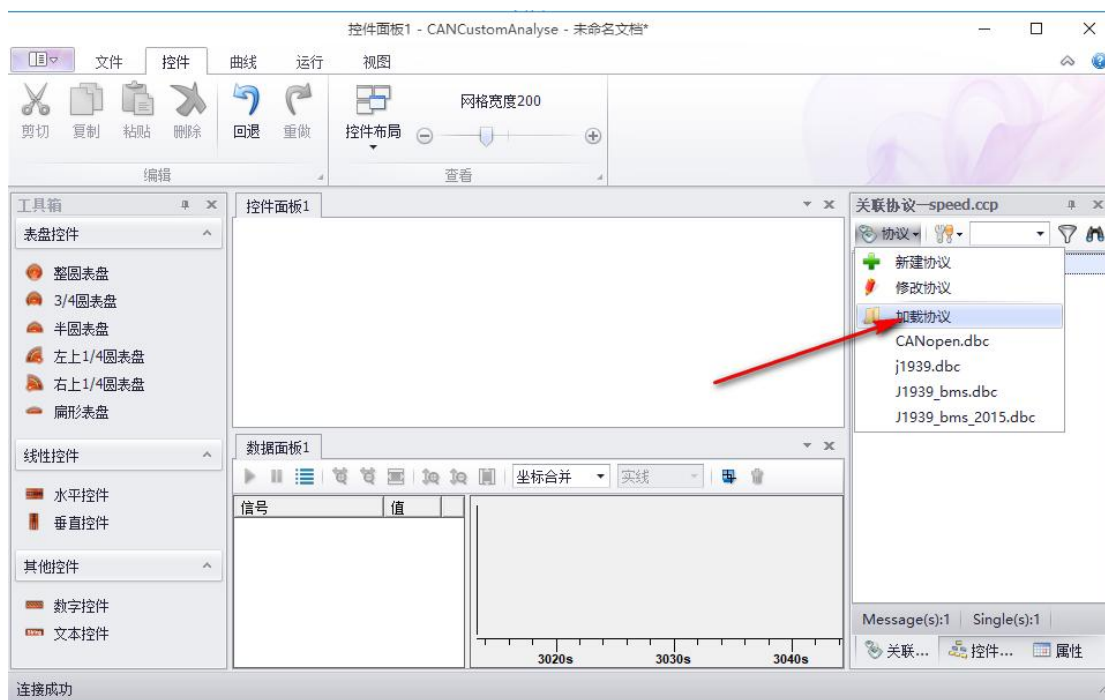
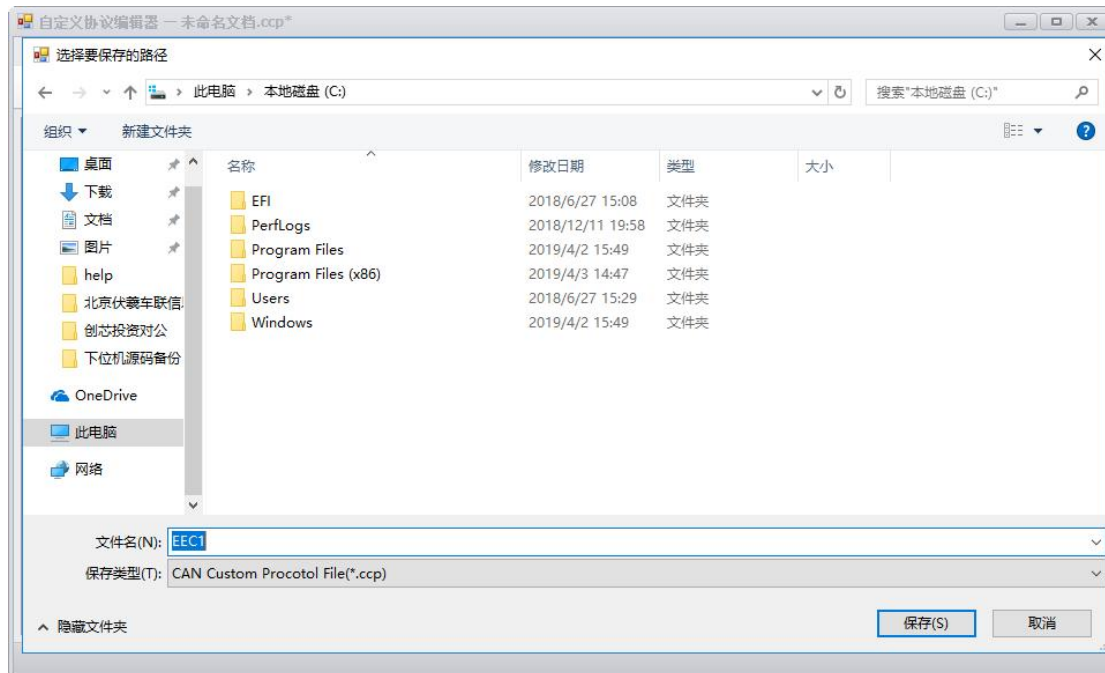


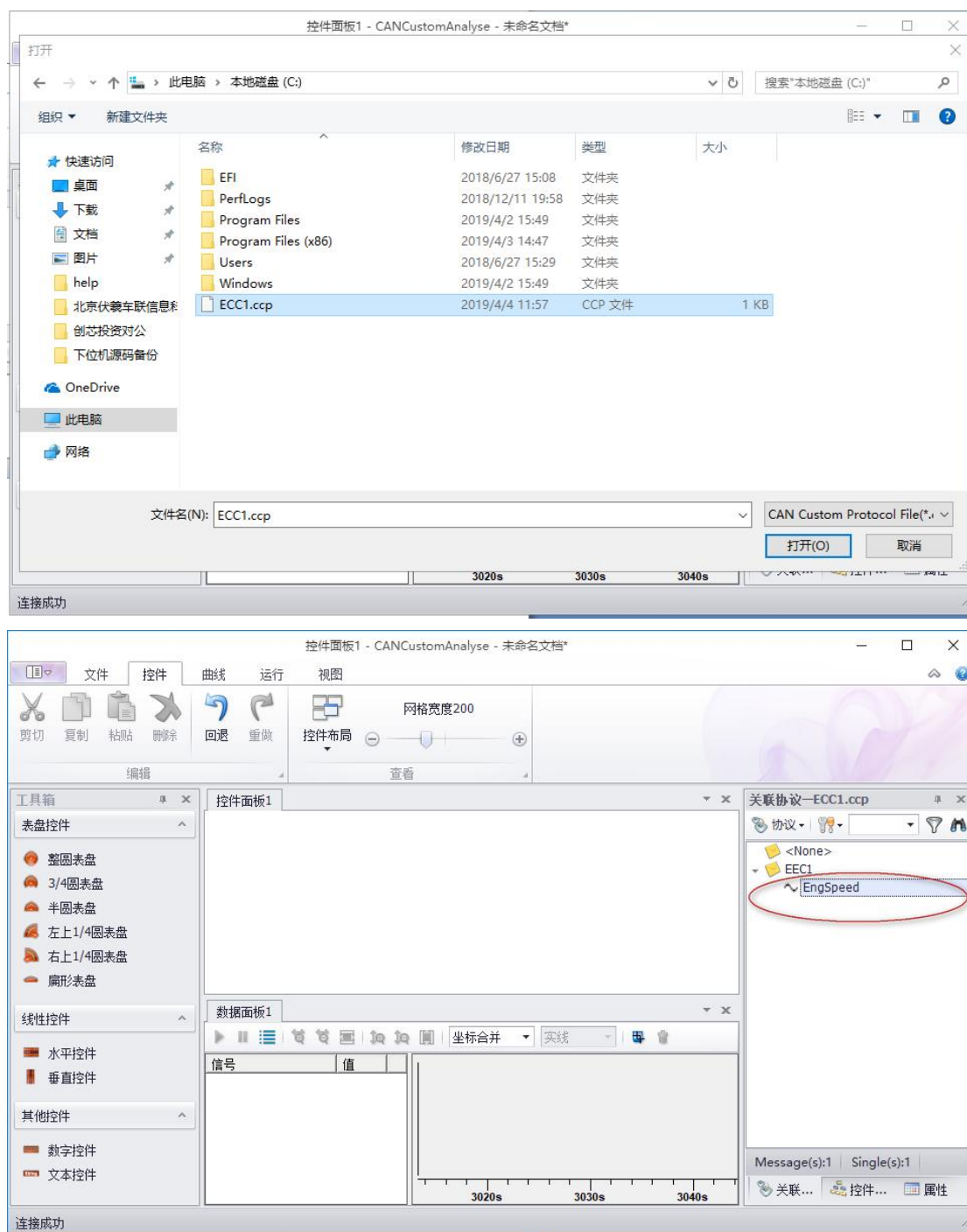


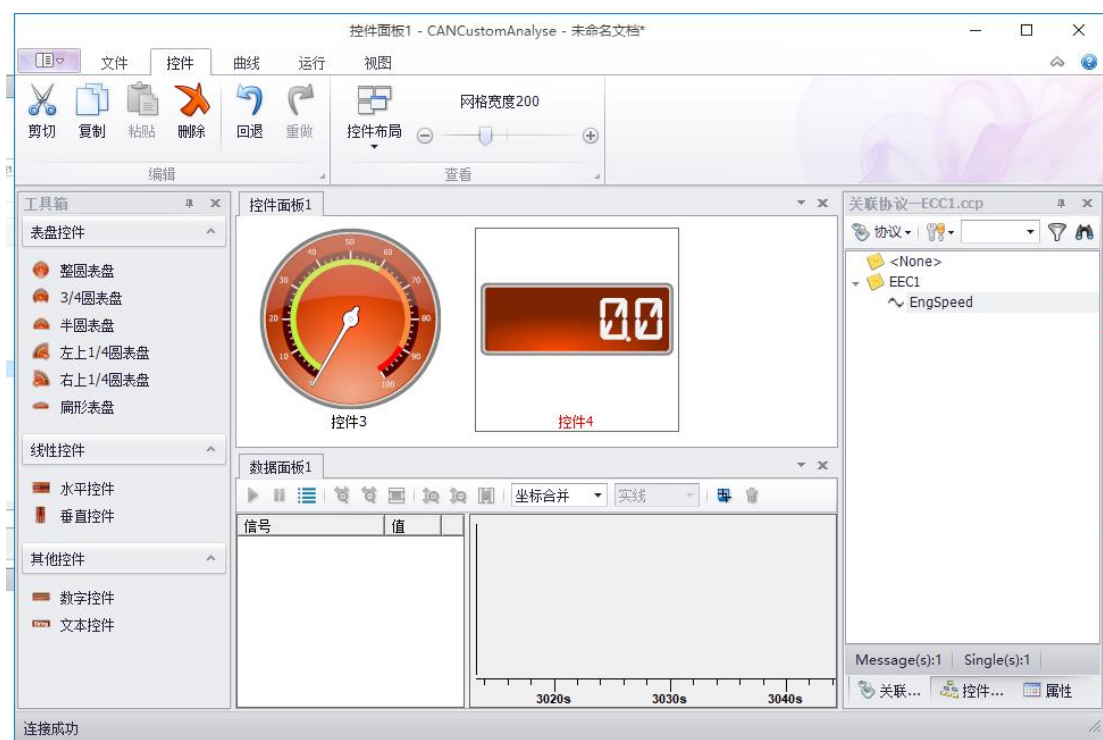
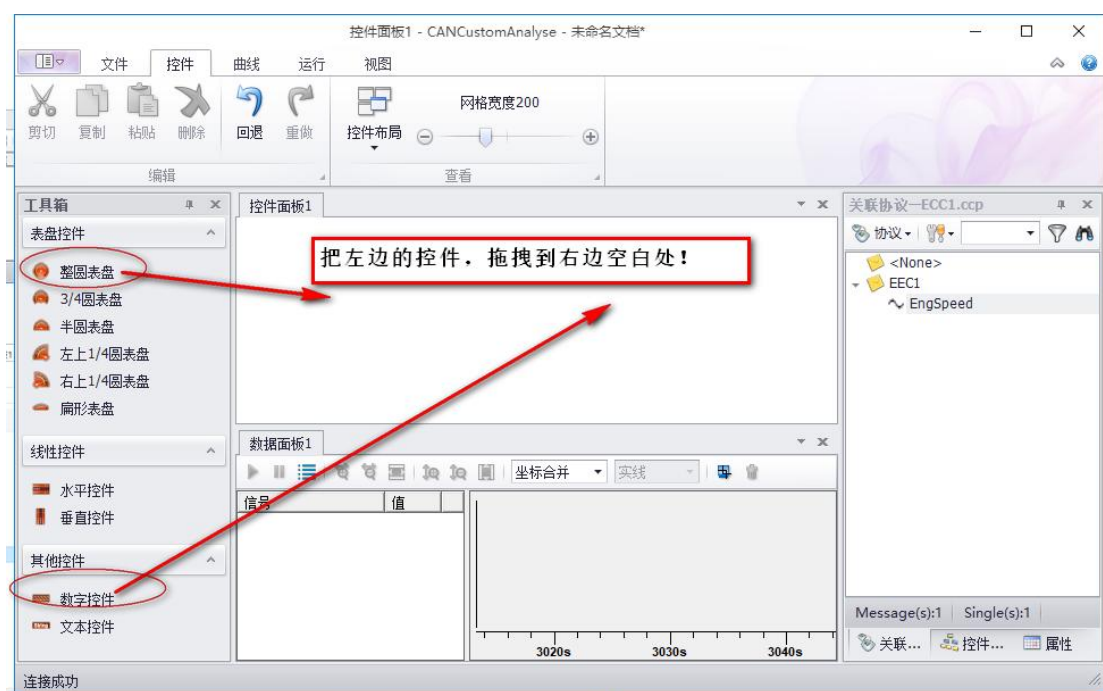


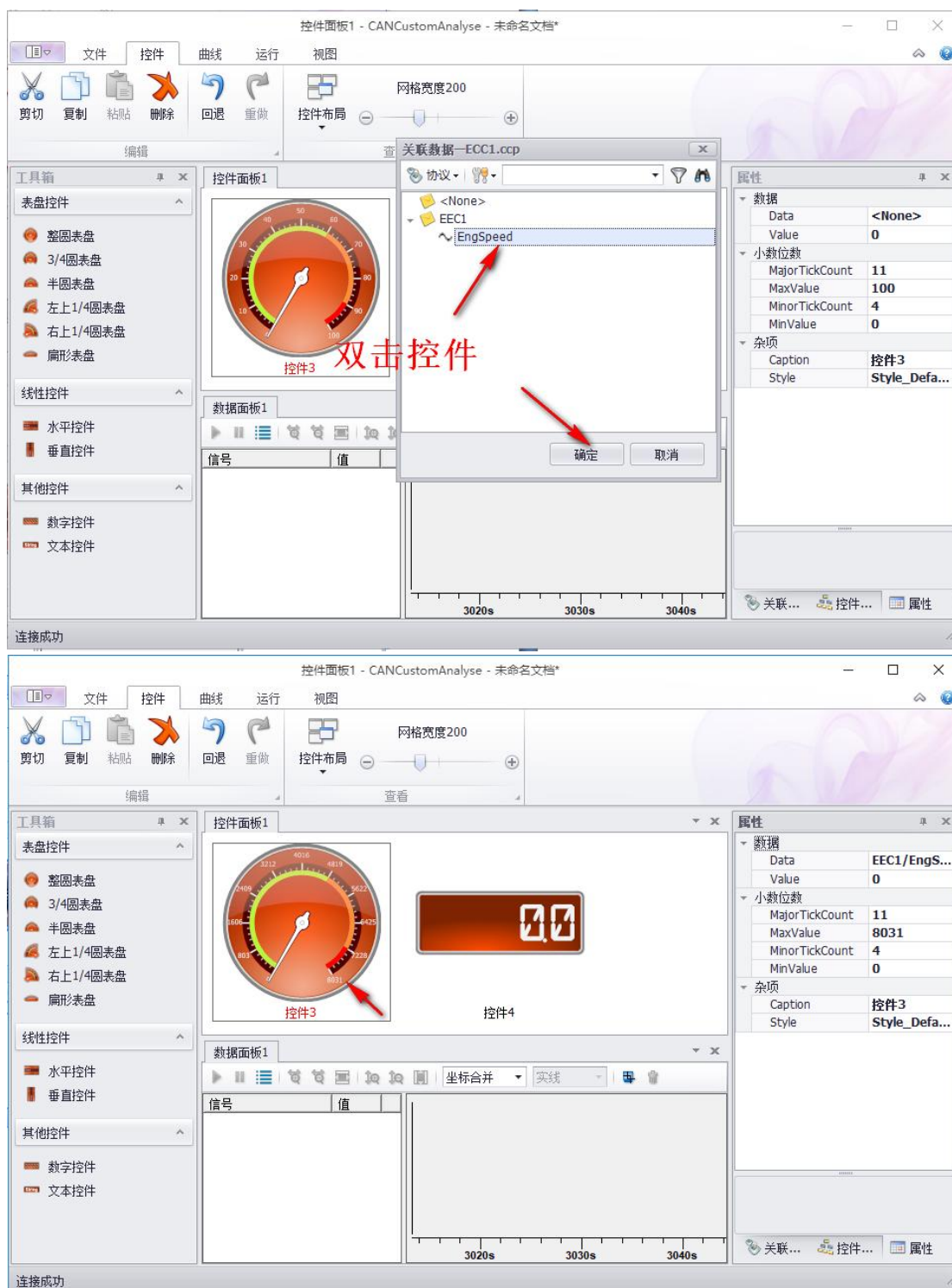
Remember: Within the signal description, there can't be Chinese, otherwise the software will flash back directly!

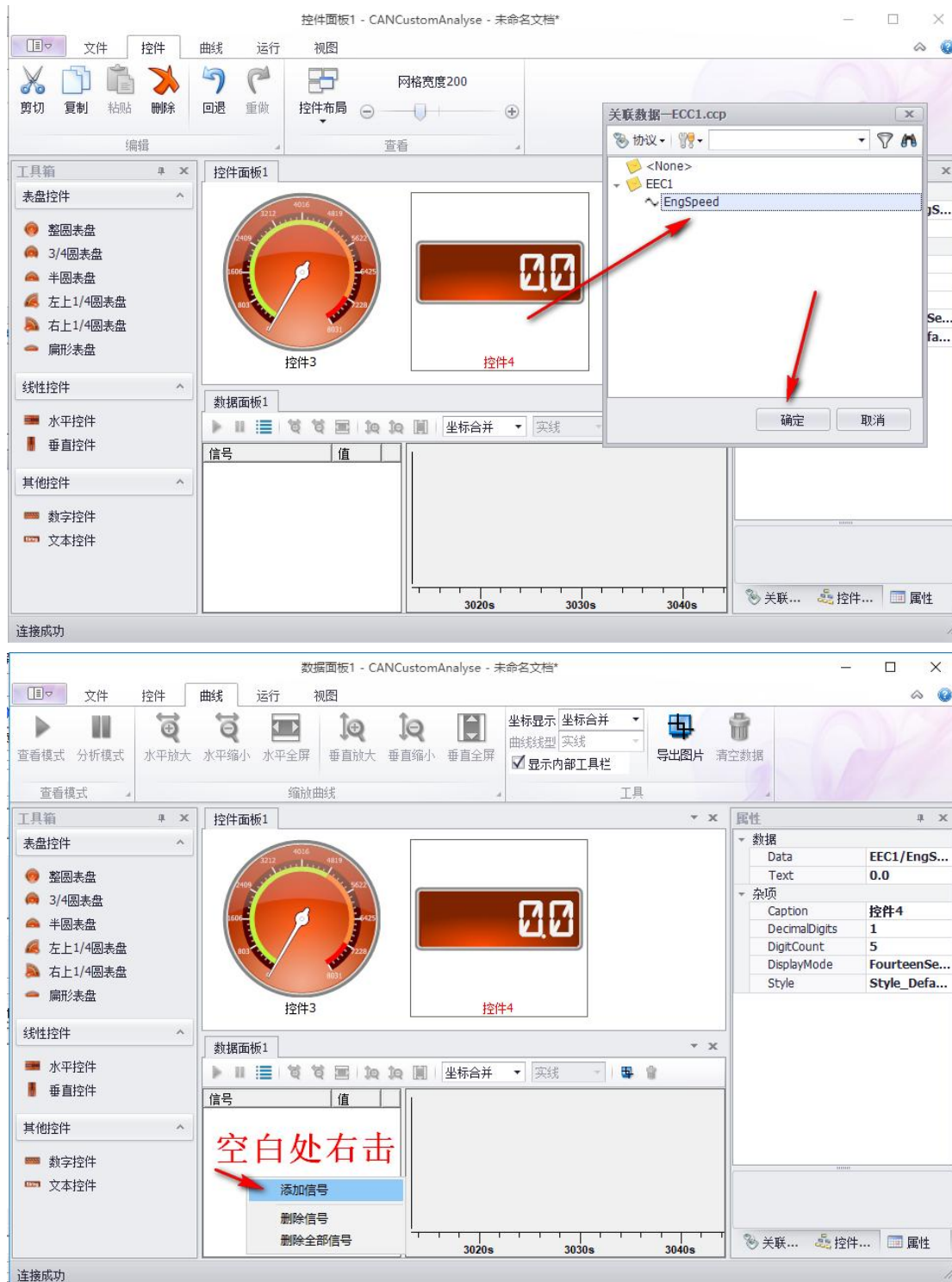


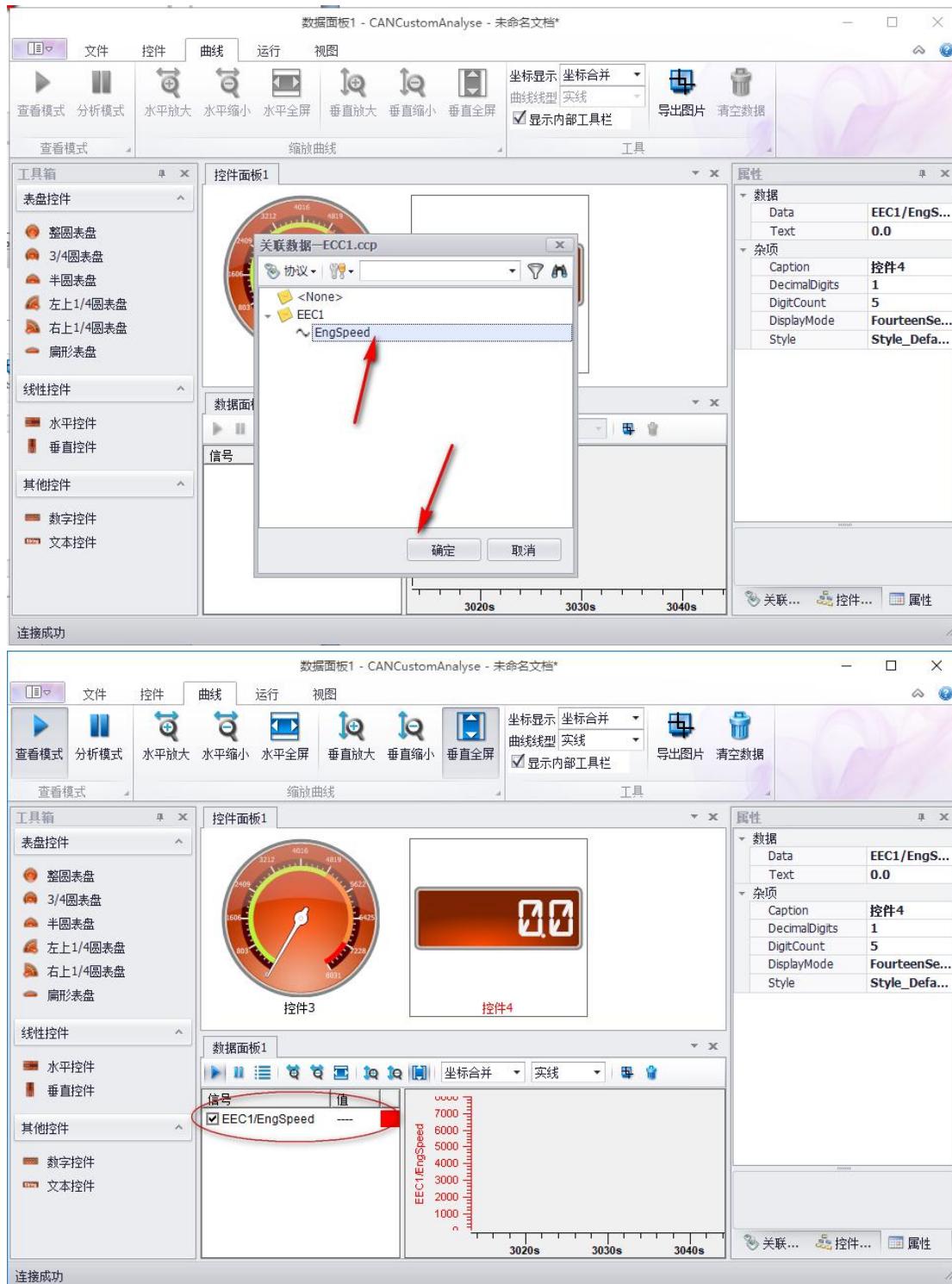


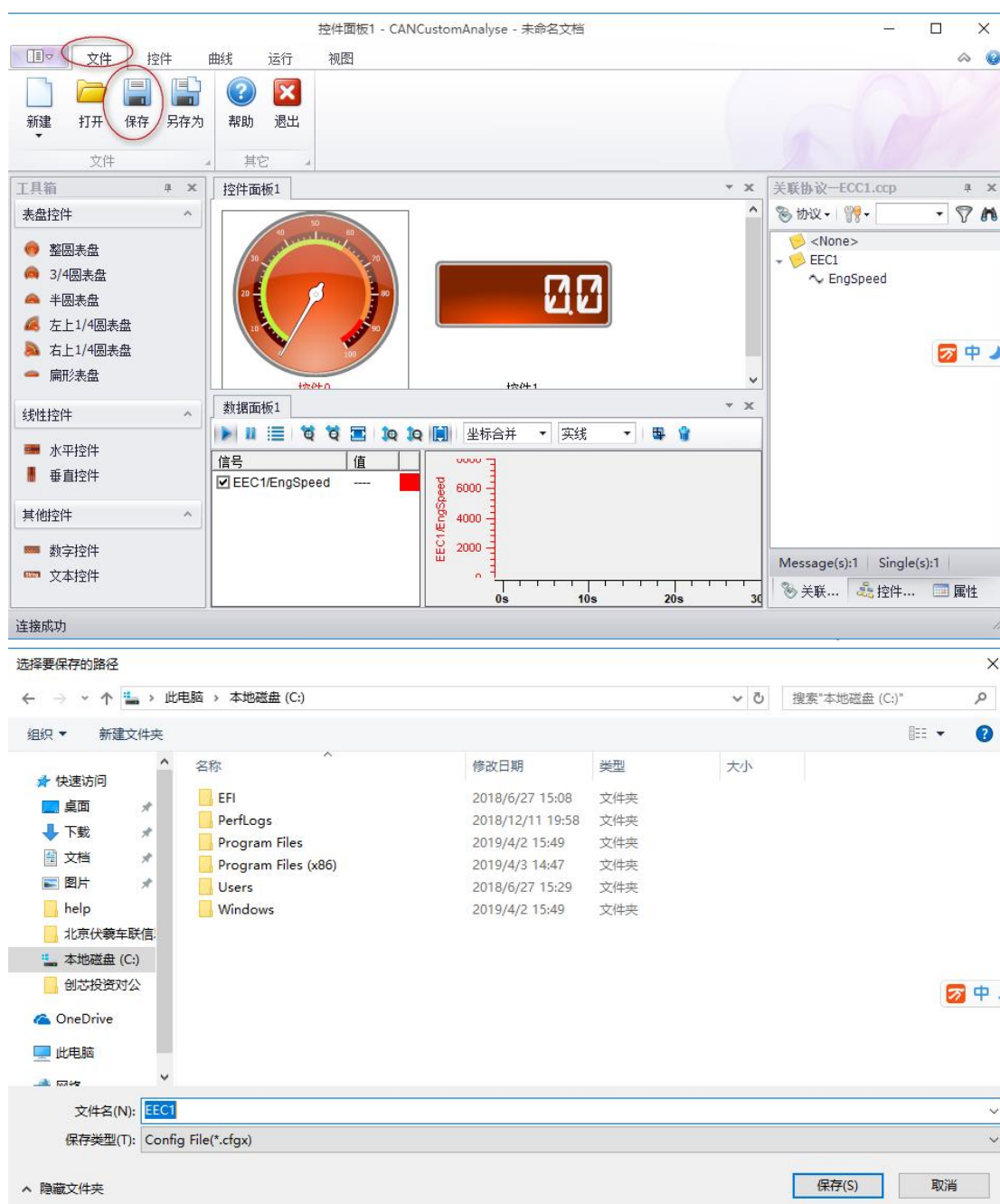










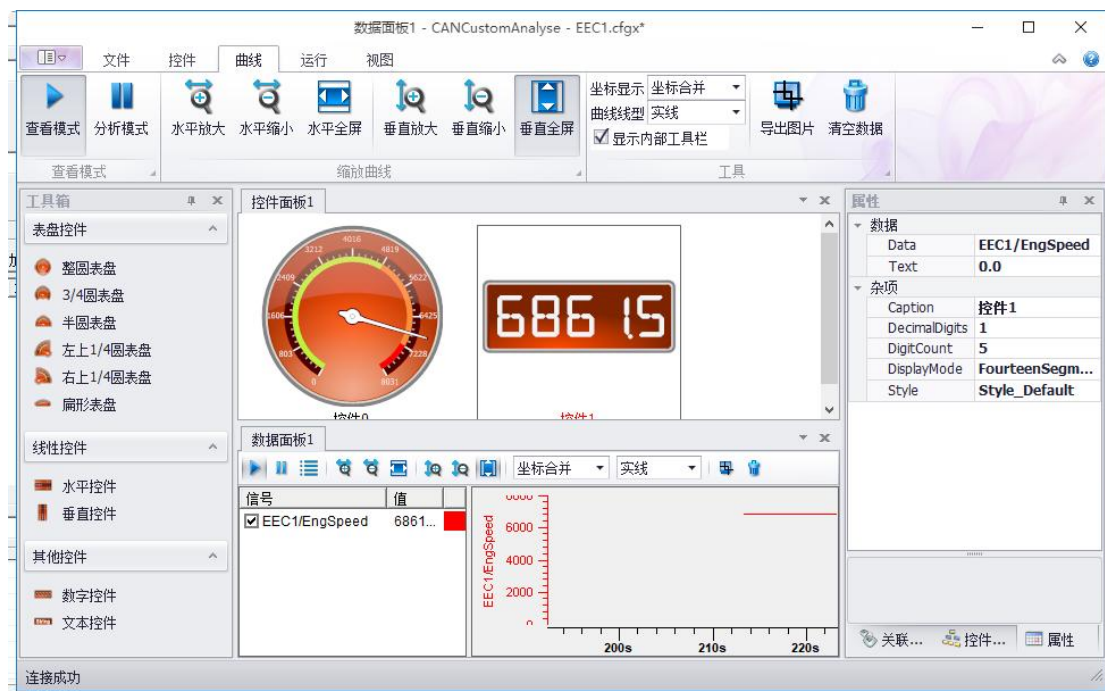


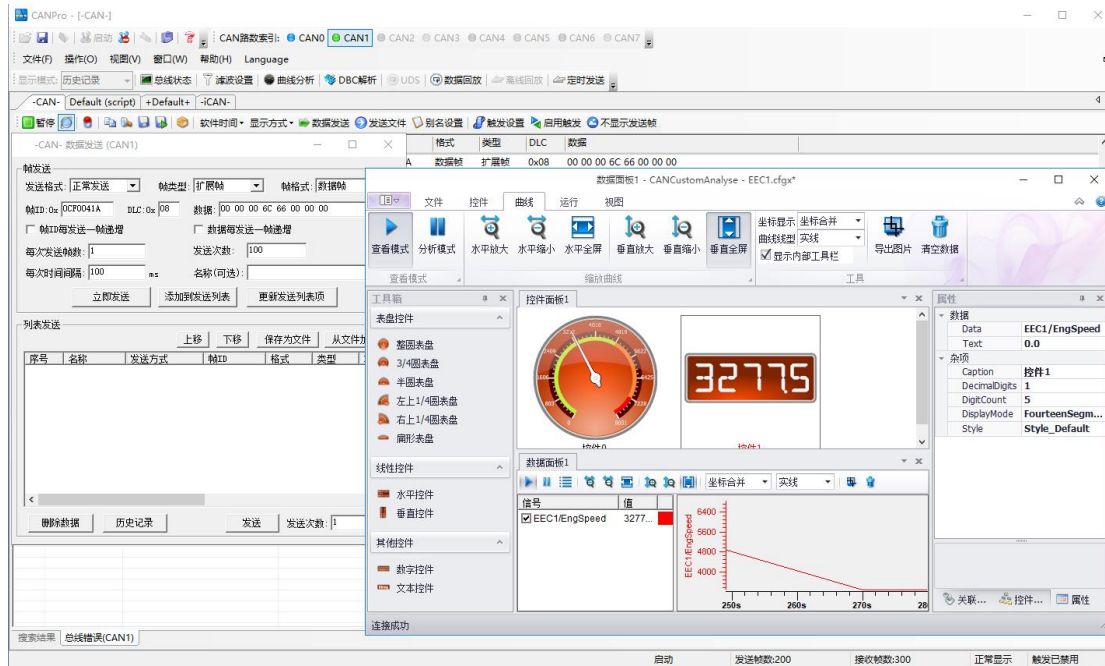
If there is data from the corresponding channel that matches the protocol, then it will be parsed.

If there is no test equipment used to send the data source, CAN1, CAN2 can be shorted up to the pair of sending, one sending and one receiving.

Wiring reference "5. Plug-in 1: USB-CAN bus adapter test.pdf".

The curve drawing above is opened in CAN0 interface, so it is used to parse the data sent and received by CAN1. The following sending needs to be done under CAN1 interface.





2、Application Scope

- Industrial Control Test
- Automotive Electronics Maintenance and Repair
- Protocol Cracking